

Tavishi Session 21

Saturday, July 11, 2026 8:36 PM

Basics

- Variables ✓
- Data Types ✓
- Type Conversion ✓
- Input and Output ✓
- Comments ✓
- Operators ✓

Control Flow

- if ✓
- if-else ✓
- if-elif-else ✓
- Nested Conditions ✓
- Match Case (optional) ✓

Loops

- for Loop ✓
- while Loop ✓
- Nested Loops ✓
- Loop Control Statements (break, continue, pass) ✓

Strings

- String Indexing
- String Slicing
- String Methods
- String Formatting

Collections

- Lists
- Tuples
- Sets
- Dictionaries

Collection Operations

- List Methods
- Tuple Operations
- Set Methods
- Dictionary Methods

Functions

- Function Definition
- Parameters and Arguments
- Return Statement
- Default Arguments
- Keyword Arguments
- Variable Length Arguments (*args, **kwargs)
- Scope (Local and Global)
- Lambda Functions
- map()
- filter()
- reduce()

String Indexing :



Train

~~~~~ "A Z A A N S U H A I L"

```
name = "A Z A A N S U H A I L"
idx → 0 1 2 3 4 5 6 7 8 9 10 11
```

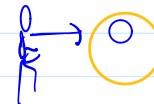
```
print(name[6])    O/P → S
```

```
name = "Azaan Suhail"
print(name[6])
```

## Why Indexing always starts with 0?

- Meaning of index is a particular is how much differ in distance from the first character.

first element  
↓  
T A V I S H I  
0 1 2 3 4 5 6



## String Slicing :

```
name = "A Z A A N S U H A I L"
      0 1 2 3 4 5 6 7 8 9 10 11
```

first name → (points to index 2)  
last name → (points to index 10)

Slicing → fetching specific part of a string

```
name[start : End : Step]
```

included (under start)  
included (under End)  
leave it for (under Step)

```
#String Slicing
print(name[0:5])
```

# String Methods:-

name = "Azaan Suhail"

#Length

`print(len(name))`

|                                                                                         |                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ✓ <code>str.count()</code>                                                              | returns the number of non-overlapping occurrences of the string.                                                                                                                |
| ✓ <code>str.endswith()</code>                                                           | returns <code>True</code> if the string ends with the suffix.                                                                                                                   |
| ✓ <code>str.startswith()</code>                                                         | returns <code>True</code> if the string starts with the prefix.                                                                                                                 |
| ✓ <code>str.join()</code>                                                               | uses the string as a delimiter for concatenating a sequence of other strings.                                                                                                   |
| ✓ <code>str.index()</code>                                                              | returns the position of the first character in the string if it was found in the string; triggers a <code>ValueError</code> if it was not found.                                |
| ✗ <code>str.find()</code>                                                               | returns the position of the first character of the first occurrence of the substring in the string; like <code>index</code> , but returns <code>-1</code> if nothing was found. |
| ✗ <code>str.rfind()</code>                                                              | Returns the position of the first character of the last occurrence of the substring in the string; returns <code>-1</code> if nothing was found.                                |
| ✓ <code>str.replace()</code>                                                            | replaces occurrences of a string with another string.                                                                                                                           |
| ✓ <code>str.strip()</code> , <code>str.rstrip()</code> ,<br>✗ <code>str.lstrip()</code> | strip spaces, including line breaks.                                                                                                                                            |
| ✓ <code>str.split()</code>                                                              | splits a string into a list of substrings using the passed separator.                                                                                                           |
| ✓ <code>str.lower()</code>                                                              | converts alphabetic characters to lower case.                                                                                                                                   |
| ✓ <code>str.upper()</code>                                                              | converts alphabetic characters to upper case.                                                                                                                                   |